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**2.** The table below gives data about some objects in our solar system.

| Object | Mass (units) | Diameter (km) | Length of day (hours) | Year length (days) | Orbital speed (km/s) | Mean temperature (°C) | Distance from Sun (units) |
|---|---|---|---|---|---|---|---|
| Mercury | 0.330 | 4 879 | 4 222.6 | 88 | 47.4 | 167 | 0.39 |
| Venus | 4.87 | 12 104 | 2 802 | 225 | 35 | 464 | 0.72 |
| Earth | 5.97 | 12 756 | 24 | 365 | 29.8 | 15 | 1.00 |
| Moon | 0.073 | 3 475 | 708.7 | | 1 | –20 | |
| Mars | 0.642 | 6 792 | 24.7 | 687 | 24.1 | –65 | 1.52 |
| Jupiter | 1 898 | 142 984 | 9.9 | 4 331 | 13.1 | –110 | 5.20 |
| Saturn | 568 | 120 536 | 10.7 | 10 747 | 9.7 | –140 | 9.54 |
| Uranus | 86.8 | 51 118 | 17.2 | 30 589 | 6.8 | –195 | 19.18 |
| Neptune | 102 | 49 528 | 16.1 | 59 800 | 5.4 | –200 | 30.06 |
| Pluto | 0.0146 | 2 370 | 163.3 | 90 560 | 4.7 | –225 | 39.53 |

(a) (i) Use the information from the table to **tick** (✓) the **two** correct statements below. [2]

Neptune is hotter than the Moon. ☐

The mean temperature of the Moon is 5 degrees less than the Earth. ☐

A year on Earth is about 4 times longer than a year on Mercury. ☐

Mercury orbits the Sun with a speed around 10 times greater than Pluto. ☐

(ii) Pluto was once considered to be a planet but is now classed as a dwarf planet. Use the data in the table to suggest a reason for the change. [1]

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(iii) Ceres is another dwarf planet and it is the only dwarf planet located in the asteroid belt. Estimate its distance from the Sun. [1]

Distance from the Sun = .................................... units

(b) Elin concludes that rocky planets with the greatest mass have the shortest day length. Explain the extent to which the data supports this conclusion. [2]

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(c) Use an equation from page 2 to calculate the velocity of the skateboarder after he catches the rugby ball. [2]

Velocity = .................................... m/s

(d) Tomos suggests that **kinetic energy** is conserved in this collision.
Explain, by using calculations, whether or not you agree.
Use an equation from page 2. [4]

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**END OF PAPER**